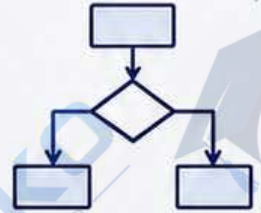


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# Sample Paper

## EXAM SAMPLE PAPER

-  Section A – Short Answer Questions (Attempt all)
-  Section B – Long Answer Questions (Attempt any two)
-  Section C – Short Answer Questions (Attempt all)
-  Section D – Long Answer Questions (Attempt any two)
-  Section E – Short Answer Questions (Attempt all)

Note: Question pattern may vary as per university guidelines.



## Semester III – Data Structure

### Sample Paper – 1 (Pattern A)

**Time: 2 Hours**

**Maximum Marks: 30**

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#### Instructions:

1. Question No. 1 is compulsory.
2. Draw neat, well-labelled diagrams wherever necessary.
3. Students may use various colours to signify answers.

**Q1.** Write answers in one or two lines and solve **any six**.

- i) What is a Data Structure?
- ii) Define an Array.
- iii) What is a Stack?
- iv) What is a Queue?
- v) Define Linked List.
- vi) What is a Binary Tree?
- vii) Define Linear Search.
- viii) What is Sorting?

**Q2.** Answer in short **any two** of the following( 3 marks each.)

- A) Explain primitive and non-primitive data structures.
- B) Differentiate between linear and non-linear data structures.
- C) Explain the concept of an array with its memory representation.
- D) Explain any three operations performed on an array.

**Q3.** Explain in detail **any one** of the following( 6 marks each.)

E) Define Stack and explain its memory representation and operations of traversing, insertion and deletion.

F) Define Queue. Explain its memory representation, operations and different types of Queue.

**Q4.** Answer in short **any two** of the following( 3 marks each.)

A) Define Linked List and state its advantages.

B) Explain different types of Linked List.

C) Define Binary Tree and explain its basic terminology.

D) Differentiate between Binary Tree and Binary Search Tree.

**Q5.** Explain in detail **any one** of the following( 6 marks each.)

E) Explain Linear Search, Binary Search and Indexed Sequential Search.

F) Explain Bubble Sort, Selection Sort and Insertion Sort with suitable examples.

## Sample Paper – 2 (Pattern B)

**Time: 2 Hours**

**Maximum Marks: 30**

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### Instructions:

1. Question No. 1 is compulsory.
2. Draw neat, well-labelled diagrams wherever necessary.
3. Students may use various colours to signify answers.

**Q1.** Write answers in one or two lines and solve **any six**.

- i) What is a non-primitive data structure?
- ii) What is meant by array traversal?
- iii) Define Stack.
- iv) What is a Circular Queue?
- v) What is a Linked List?
- vi) Define Complete Binary Tree.
- vii) What is Indexed Sequential Search?
- viii) Define Selection Sort.

**Q2.** Explain in detail **any one** of the following( 6 marks each.)

**E)** Explain primitive and non-primitive data structures. Also differentiate between linear and non-linear data structures with suitable examples.

**F)** Define an array. Explain its memory representation and describe the operations of insertion and deletion with suitable examples.

**Q3.** Answer in short **any two** of the following( 3 marks each.)

- A) Explain the concept of Stack and mention its applications.
- B) Describe the memory representation of a Queue.
- C) Differentiate between Stack and Queue.
- D) Explain any three types of Queue.

**Q4.** Explain in detail **any one** of the following( 6 marks each.)

**E)** Define Linked List. Explain its memory representation and different types of Linked Lists with suitable diagrams.

**F)** Explain Binary Search Tree and its important terminologies. Describe insertion and deletion operations in a Binary Search Tree.

**Q5.** Answer in short **any two** of the following( 3 marks each.)

**A)** Explain the working principle of Linear Search.

**B)** What is Binary Search? State its basic requirement.

**C)** Explain the concept of Bubble Sort.

**D)** Differentiate between Bubble Sort and Selection Sort.

## Semester III – RDBMS

### Sample Paper – 1 (Pattern A)

**Time: 2 Hours**

**Maximum Marks: 30**

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**Instructions:**

- 4. Question No. 1 is compulsory.**
- 5. Draw neat, well-labelled diagrams wherever necessary.**
- 6. Students may use various colours to signify answers.**

**Q1.** Write answers in one or two lines and solve **any six**.

- i) Define Database.
- ii) What is a Relational Model?
- iii) What is SQL?
- iv) Define View.
- v) What is an Index?
- vi) What is PL/SQL?
- vii) Define Trigger.
- viii) What is Database Backup?

**Q2.** Answer in short **any two** of the following( 3 marks each.)

- A)** Differentiate between Data and Information.
- B)** Explain any three characteristics of a Database.
- C)** Explain the advantages of using a Database.
- D)** Differentiate between DDL and DML commands with suitable examples.

**Q3.** Explain in detail **any one** of the following( 6 marks each.)

**E)** Explain different types of Data Constraints. Describe the operations performed on a table using constraints with suitable examples.

**F)** Explain Views and Indexes in detail. Discuss their need, advantages and limitations. Also explain the Union of Tables.

**Q4.** Answer in short **any two** of the following( 3 marks each.)

**A)** What is PL/SQL? State its advantages.

**B)** Explain the basic block structure of PL/SQL.

**C)** What is Exception Handling in PL/SQL?

**D)** Define Cursor. State its types.

**Q5.** Explain in detail **any one** of the following( 6 marks each.)

**E)** Explain Procedure and Function in PL/SQL. Differentiate between a Procedure and a Function with suitable examples.

**F)** Explain Backup and Recovery in a Database. Discuss Archive Logs and Transaction Control in detail.

## Sample Paper – 2 (Pattern B)

**Time: 2 Hours**

**Maximum Marks: 30**

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### Instructions:

4. **Question No. 1 is compulsory.**
5. **Draw neat, well-labelled diagrams wherever necessary.**
6. **Students may use various colours to signify answers.**

**Q1.** Write answers in one or two lines and solve **any six**.

- i) What is a Database Management System?
- ii) Define Database Architecture.
- iii) What is the HAVING clause in SQL?
- iv) Define Data Constraint.
- v) What is a View?
- vi) What is an Exception in PL/SQL?
- vii) Define a Cursor.
- viii) What is Transaction Control?

**Q2.** Explain in detail **any one** of the following( 6 marks each.)

**E)** Explain the characteristics and needs of a Database. Describe the three components of DBMS Architecture: **DBA, DB and Role**.

**F)** Explain SQL in detail. Discuss its features, characteristics and advantages. Explain **DDL, DML, DCL and TCL commands** with suitable examples.

**Q3.** Answer in short **any two** of the following( 3 marks each.)

- A)** Explain any three types of Data Constraints.
- B)** What is a View? State its needs and advantages.
- C)** Explain the need and advantages of an Index.
- D)** Write a short note on Character Functions and Number Functions in SQL.

**Q4.** Explain in detail **any one** of the following( 6 marks each.)

**E)** Explain the structure of a PL/SQL block. Discuss different data types and exception handling in PL/SQL.

**F)** Explain Cursor in PL/SQL. Describe the types of Cursor, attributes of Cursor and the life cycle of a Cursor.

**Q5.** Answer in short **any two** of the following( 3 marks each.)

**A)** Define Procedure and state its advantages.

**B)** Differentiate between Procedure and Function.

**C)** What is a Trigger? Mention its importance.

**D)** Write a short note on Database Recovery and Transaction Control.

## Semester III – IKS

### Sample Paper – 1 (Pattern A)

**Time: 2 Hours**

**Maximum Marks: 30**

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**Instructions:**

- 7. Question No. 1 is compulsory.**
- 8. Draw neat, well-labelled diagrams wherever necessary.**
- 9. Students may use various colours to signify answers.**

**Q1.** Write answers in one or two lines and solve **any six**.

- i) What is meant by Indian Knowledge System (IKS)?
- ii) What is Corpus-A Classification Framework?
- iii) Name any two languages associated with Indian Knowledge Systems.
- iv) What is the significance of zero in Indian mathematics?
- v) Define a numeral system.
- vi) What is a square root?
- vii) What is the value of  $\pi$  (Pi)?
- viii) What is meant by Binary Mathematics?

**Q2.** Answer in short **any two** of the following( 3 marks each.)

- A)** Explain the importance of ancient knowledge in the Indian Knowledge System.
- B)** Define Indian Knowledge System (IKS) and explain its major aspects.
- C)** Explain the Corpus-A Classification Framework of IKS.
- D)** Write a short note on the history of IKS.

**Q3.** Explain in detail **any one** of the following( 6 marks each.)

**E)** Explain the computational concepts found in **Ashtadhyayi** and discuss the role of Sanskrit in Natural Language Processing.

**F)** Explain the contribution of Indian mathematicians to mathematics. Discuss the concept of algebra, arithmetic and geometry in Indian mathematical tradition.

**Q4.** Answer in short **any two** of the following( 3 marks each.)

**A)** Explain the historical evidence of the Indian Number System.

**B)** Write a note on the salient features of Indian Numerals.

**C)** Explain the concept and importance of Zero in Indian mathematics.

**D)** Differentiate between large numbers and their representation.

**Q5.** Explain in detail **any one** of the following( 6 marks each.)

**E)** Explain the ancient Indian methods of measurement for **Time, Distance and Weight** with suitable examples.

**F)** Explain **Arithmetic, Square Root, Series and Progressions** in Indian mathematics. Also discuss the value of  $\pi$  (Pi).

## Sample Paper – 2 (Pattern B)

Time: 2 Hours

Maximum Marks: 30

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### Instructions:

7. Question No. 1 is compulsory.
8. Draw neat, well-labelled diagrams wherever necessary.
9. Students may use various colours to signify answers.

**Q1.** Write answers in one or two lines and solve **any six**.

- i) What is meant by Ancient Indian Knowledge?
- ii) What is Caturdasa Vidyasthana?
- iii) Name any two aspects of the Indian Knowledge System.
- iv) What is the importance of the decimal number system?
- v) Define Zero.
- vi) What is a Binary Number System?
- vii) Define Series and Progressions.
- viii) What is meant by Indian measures of distance?

**Q2.** Explain in detail **any one** of the following( 6 marks each.)

**E)** Explain the importance of Ancient Knowledge and discuss the definition, significance and major aspects of the Indian Knowledge System.

**F)** Explain the **Caturdasa Vidyasthana** and discuss the history and unique aspects of the Indian Knowledge System.

**Q3.** Answer in short **any two** of the following( 3 marks each.)

- A) Explain the computational concept in Ashtadhyayi.
- B) Write a short note on the role of Sanskrit in Natural Language Processing.
- C) Explain the contribution of Indian mathematicians to mathematics.
- D) Write a short note on the Indian concepts of algebra, arithmetic and geometry.

**Q4.** Explain in detail **any one** of the following( 6 marks each.)

E) Explain the historical evidence and salient features of the Indian Number System. Discuss the development of Indian numerals.

F) Explain the concept of Zero and its importance. Discuss large numbers, their representation and the place value system.

**Q5.** Answer in short **any two** of the following( 3 marks each.)

A) Explain the unique aspects of Indian measurements for Time, Distance and Weight.

B) Write a short note on Arithmetic in Indian mathematics.

C) Explain the concept of Square Root and its significance.

D) Write a short note on Algebra and Binary Mathematics in India.